

Electric Vehicles

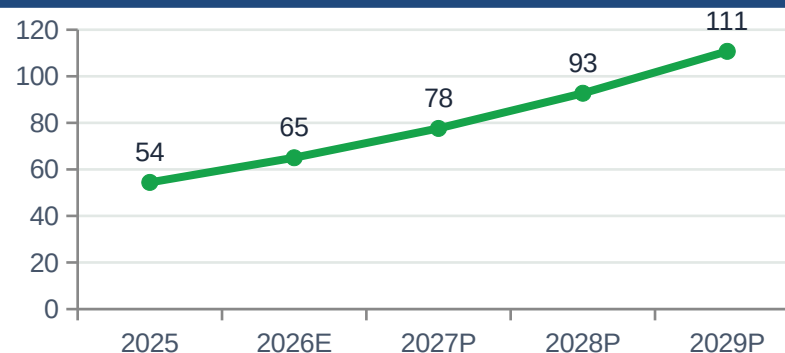
Industry Size ([Link](#))

US\$65 Bn

FY2026

- 19.44% CAGR (2025-29P) •
- Two-wheelers ~92% of EV volumes; passenger EVs the fastest-growing segment

Industry Growth ([link](#))



According to Mordor Intelligence, the India EV market is estimated at US\$54.4 billion in 2025 and is projected to reach US\$65.0 billion in FY2026, growing to US\$110.7 billion by 2029 at a CAGR of 19.44%. Growth is led by government incentive schemes, falling battery costs and a fast-expanding charging network.

Key players ([Link](#))

- TVS Motor – 25%
- Bajaj Auto – 22%
- Ather Energy – 16%
- Hero Vida – 11%
- Ola Electric – 8%

Business Model and Industry Product Mix

Key Segments ([Link](#))

Two-Wheelers	~92%	Dominant – affordability, low running cost
Passenger Vehicles	~5%	Fastest growing; SUVs lead new launches
Three-Wheelers	~2%	High EV penetration in cargo/passenger use
Buses & Commercial	<1%	Scaling via PM E-DRIVE / state fleets

SNAPSHOT

- India's largest EV brand by segment: Tata Motors (4W) and TVS Motor (2W)
- Electric 2W penetration crossed 10% of total 2W sales for the first time in June 2026
- Passenger EV penetration reached ~4.2-4.9% of PV sales in FY26

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Value Chain ([Link](#))

1. Critical Minerals
(Li, Co, Ni, REE)



2. Battery Cells
& Packs (LFP)



3. EV Motor &
Power Electronics



4. Vehicle
Assembly (OEM)



5. Charging &
After-sales

Porter's 5 Forces ([Link](#))

Threat of New Entrants
Moderate

Govt. incentives lower entry barriers; Vinfact, BYD, Hyundai entering. But high capex, R&D and scale economics favour incumbents like Tata, Mahindra.

Bargaining Power of Buyers
High

Consumers can choose across brands (Tata, Mahindra, MG, TVS, Bajaj); price-sensitive; fleet/government buyers command extra leverage.

Bargaining Power of Suppliers
Mod-High

Battery cell and rare-earth magnet supply concentrated with China-linked sources; vertical integration (Tata Agratas, Ola Cells) is reducing this power.

Threat of Substitutes
Moderate

ICE vehicles, CNG, hybrids and public transit/micromobility remain viable alternatives, especially outside metros.

Industry Rivalry
Very High

Rapid share shifts (Tata vs Mahindra vs MG in PVs; TVS/Bajaj/Ather vs Ola in 2W), frequent launches and price competition.

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Growth Drivers ([Link](#))

- PM E-DRIVE scheme (₹10,900 cr) extended to March 2028, backing e-buses, e-trucks and charging infrastructure (source)
- PLI-ACC scheme targeting 50 GWh of domestic battery-cell capacity to cut import dependence (source)
- Falling battery costs and customs-duty exemptions on lithium-ion inputs (cobalt, lithium, nickel) (source)
- Record FY26 EV retail momentum: H1 2026 passenger EV sales up 79% YoY; electric 2W penetration crossed 10% for the first time in June 2026 (source)

Emerging Trends ([Link](#))

- LFP chemistry now dominant in new battery packs across 2W, 3W and mass-market PVs (source)
- Government's draft 'Battery Aadhaar' rules (Jan 2026) introduce unique lifecycle codes for traceability and recycling (source)
- Battery-as-a-service / swapping models emerging to reduce upfront ownership cost (source)
- Consolidation in electric 2W: legacy OEMs (TVS, Bajaj, Hero) gaining share from pure-play Ola Electric, which has ceded leadership (source)

Recent Investments/M&A ([Link](#))

- Tata Group's Agravas signed a ~\$1.6 Bn (₹13,000 cr) deal for a lithium-ion cell gigafactory at Sanand (Gujarat); initial 20 GWh, expandable to 40 GWh (source)
- Ola Electric approved a ₹2,000 cr investment (May 2026) into its EV and battery-cell subsidiaries (OET, OCT), and is separately raising up to ₹2,000 cr via a stake sale in its battery arm (source)
- Reliance Industries is building a ~30 GWh battery gigafactory (~₹7,500 cr investment) (source)
- Samvardhana Motherson entered exclusive talks to acquire Nexans' wiring-harness business for €207 mn, strengthening EV component capability (source)

Key Business Risks ([Link](#))

- ~80-90% import dependence on China for rare-earth magnets and critical battery minerals; April 2025 export curbs exposed this vulnerability (source)
- PM E-DRIVE FY27 outlay for e-2W/e-3W demand incentives cut ~62.5% as subsidies phase out by March 2026, testing affordability (source)
- Inverted GST/duty structure on components raises the cost of local EV manufacturing versus finished imports (source)
- Execution and financial-stress risk at pure-play EV makers (e.g., Ola Electric's continuing losses and market-share erosion) (source)

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Balance Sheet ([Link](#))

Financial Snapshot FY26

Metric	Ola Electric*	Tata Motors EV
Revenue	₹2,253 cr	Not disclosed separately
Net Profit / (Loss)	(₹1,833) cr	n.a. (part of PV segment)
Market Cap	~₹17,900 cr	n.a. (listed as Tata Motors)
Price / Book	~5.1x	n.a.
Planned EV Capex	Gigafactory: ₹3,500 cr	₹16,000–18,000 cr by FY30

Income Statement & Valuation ([Link](#))

Valuation Approach:

- Loss-making, pure-play EV OEMs (e.g., Ola Electric) are typically valued on EV/Sales and Price-to-Book, given negative earnings
- Diversified OEMs (Tata Motors, Mahindra) are valued on consolidated forward P/E and EV/EBITDA; their EV business is priced as an embedded growth option rather than standalone
- DCF is used selectively for gigafactory / battery investments with long-dated, capital-intensive cash flow profiles

Unit Economics ([Link](#))

Parameter	Electric Vehicle	ICE Vehicle
Upfront price premium	+15-25% vs ICE	Baseline
Running cost / km	~80% lower	Baseline
5-yr total cost of ownership*	Lower by ~₹3.6 lakh	Higher
Largest cost component	Battery (~35-40% of cost)	Engine & drivetrain

**Illustrative, for a typical mid-size passenger EV vs comparable petrol car over 5 years / ~75,000 km.*

Key Business Risks

- High Battery Costs** Batteries account for **30–40% of an EV's total cost**, making manufacturers highly vulnerable to fluctuations in the prices of lithium, cobalt, nickel, and other critical minerals.
- Raw Material Supply Risk** The EV industry relies heavily on imported critical minerals. Geopolitical tensions, export restrictions, and supply shortages can disrupt production and increase costs.
- Rapid Technological Advancements** The industry is evolving quickly with new battery technologies, AI, and autonomous driving features.
- Intense Market Competition** Competition from global automakers, new EV startups, and low-cost Chinese manufacturers has increased pricing pressure and reduced profit margins.